Re-evaluating Swiss reference dates and studying phenological shifts using four independent regional monitoring datasets

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overview of the datasets

Table 1: Overview of the five component analyses.

Region	Years	Species	Classification
Bavaria, DE	2020–2024	116	Territory count > 0
Switzerland	2021–2025	$1\,\mathrm{km}^2$ grid	AC ≥ 4 and < 20, or $= 50$; $5\,\mathrm{km}$ buffer
Switzerland	2007–2025	$1\,\mathrm{km}^2$ grid	AC ≥ 4 and < 20, or $= 50$; $5\,\mathrm{km}$ buffer
Baden-Württemberg, DE	2018–2023	303	AC, pooled across years
Baden-Württemberg, DE (Ornitho)	2020–2025	134	B/C prefix

questions we are trying to address:

1. how does breeder phenology compare to reference dates?

2. how does migrant phenology compare to reference dates?

3. have breeder and migrant peak dates shifted over years?

4. are the responses to aforementioned questions consistent across datasets?

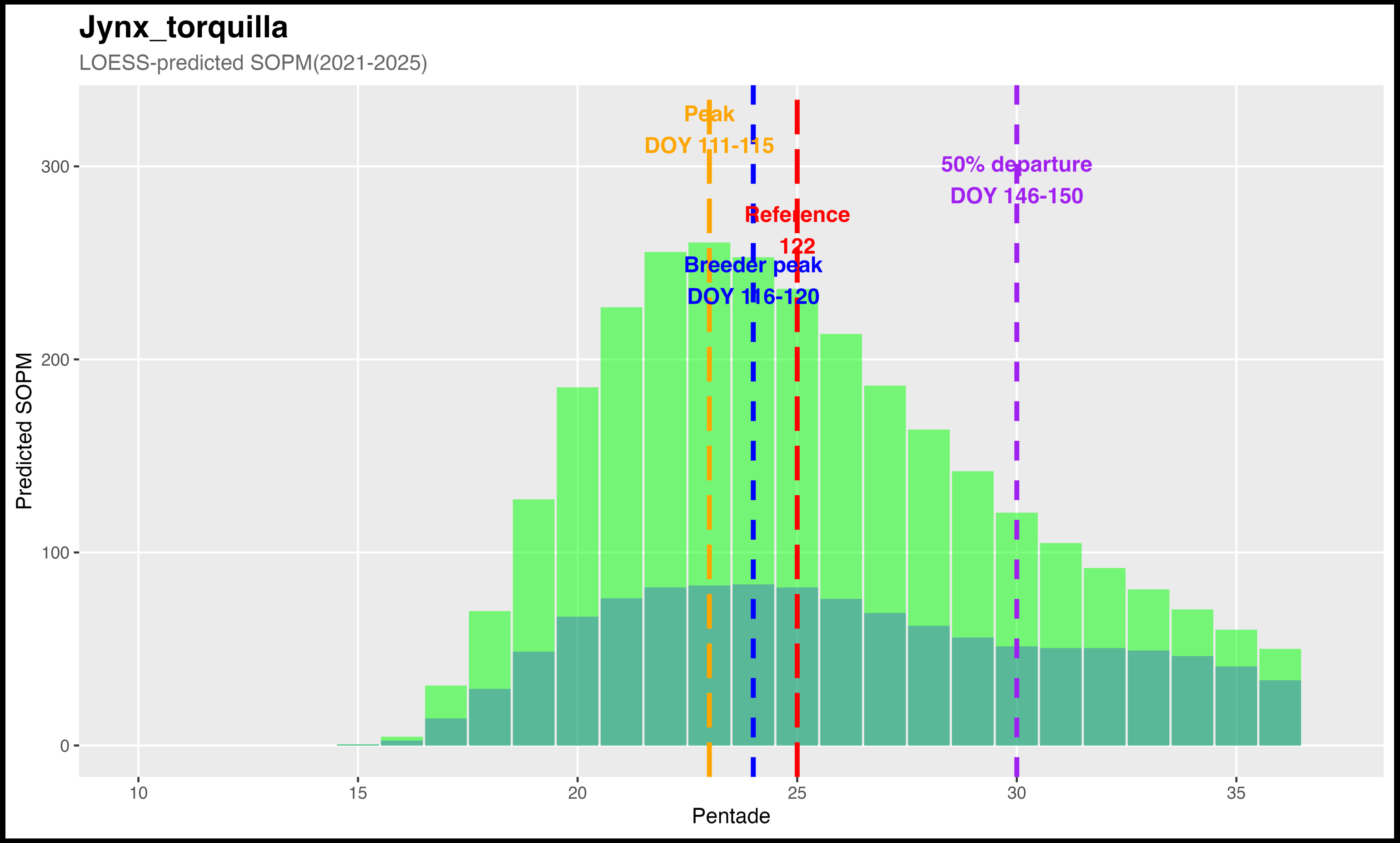
metrics

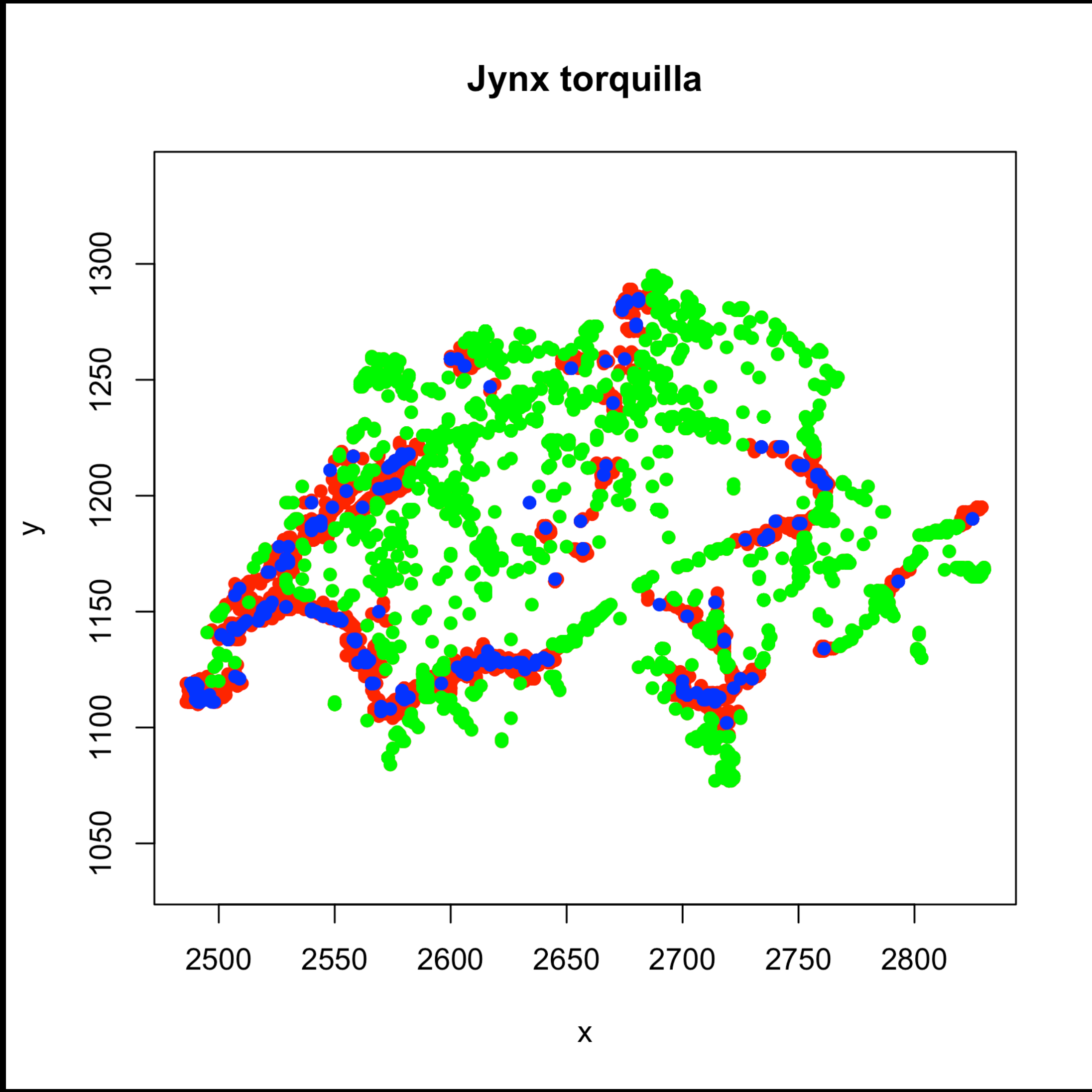
Table 2: Phenological metrics extracted from the LOESS curves.

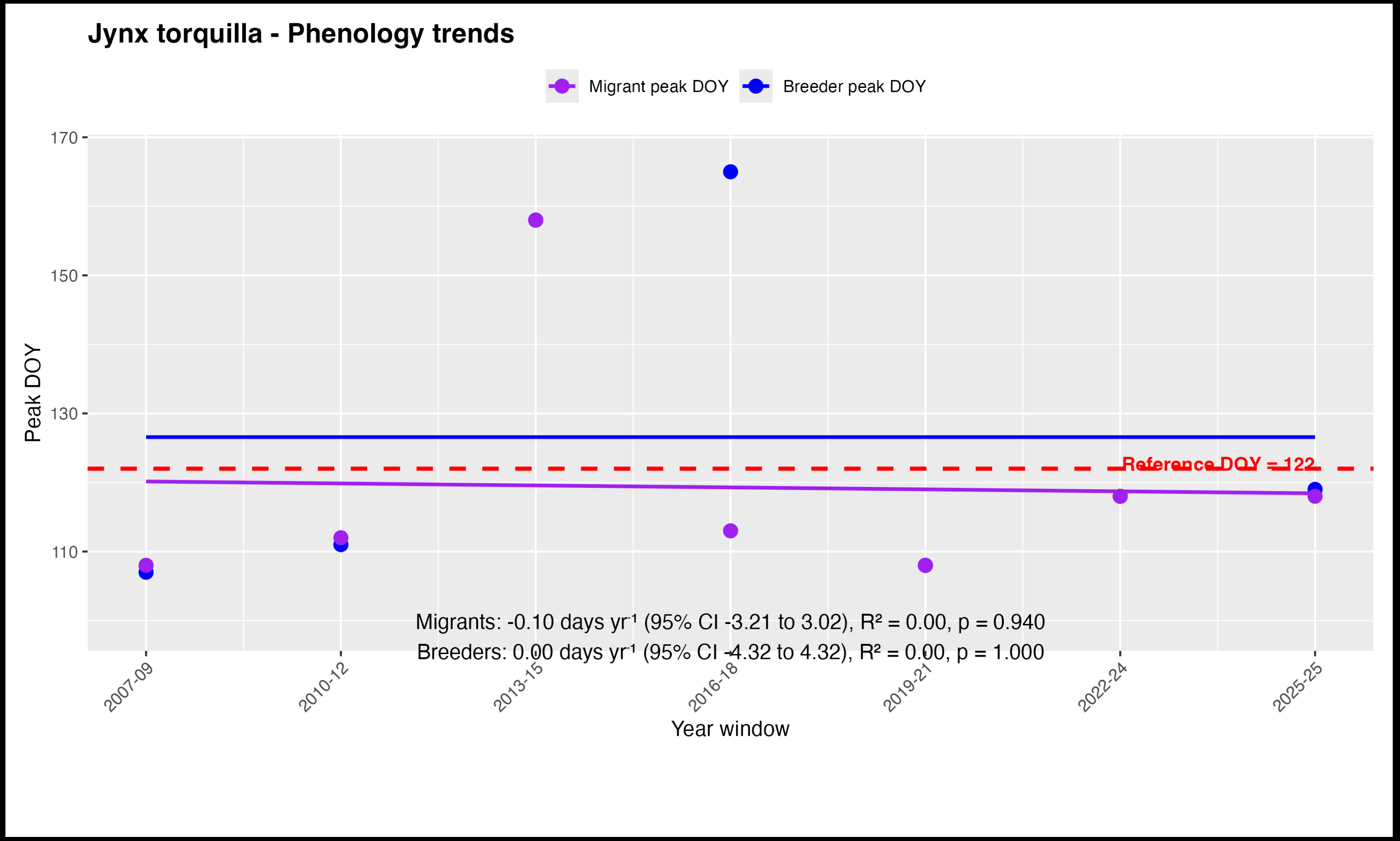
Metric	Ecological indication
`Breeder_Peak_DOY`	DOY at which the `breeding_sites` curve reaches its maxima
`Migrant_Peak_DOY`	DOY at which the `non_breeding_sites` curve reaches its maxima
`Reference_DOY − Breeder_peak_doy`	negative response means that the breeding population peak has arrived after the reference date and vice versa
`Reference_DOY − Migrant_peak_doy`	positive response indicates that the non-breeding population peak before the reference date and vice versa

long distance migrants: *Jynx torquilla*

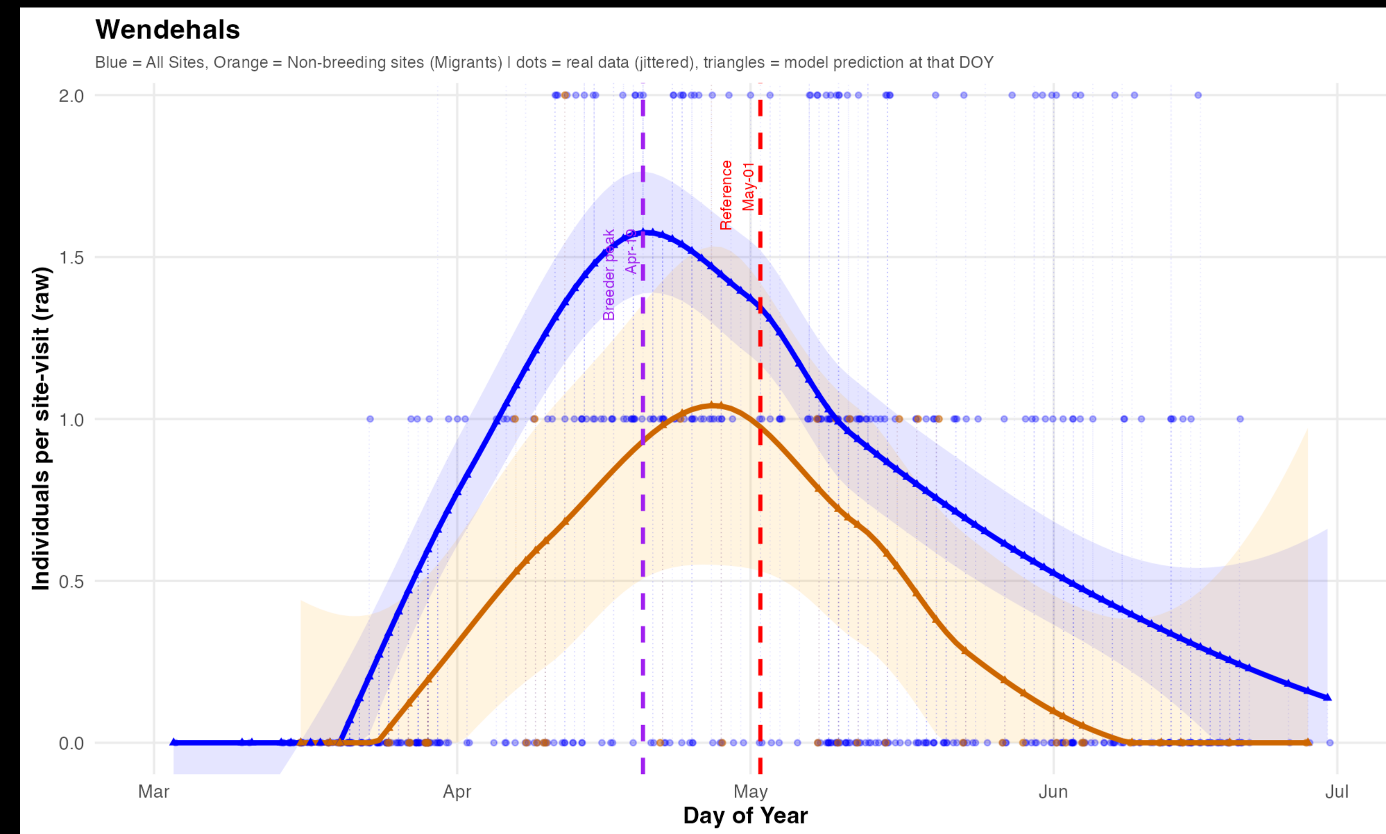
swiss dataset > distribution of sites, subpopulations and phenological trends

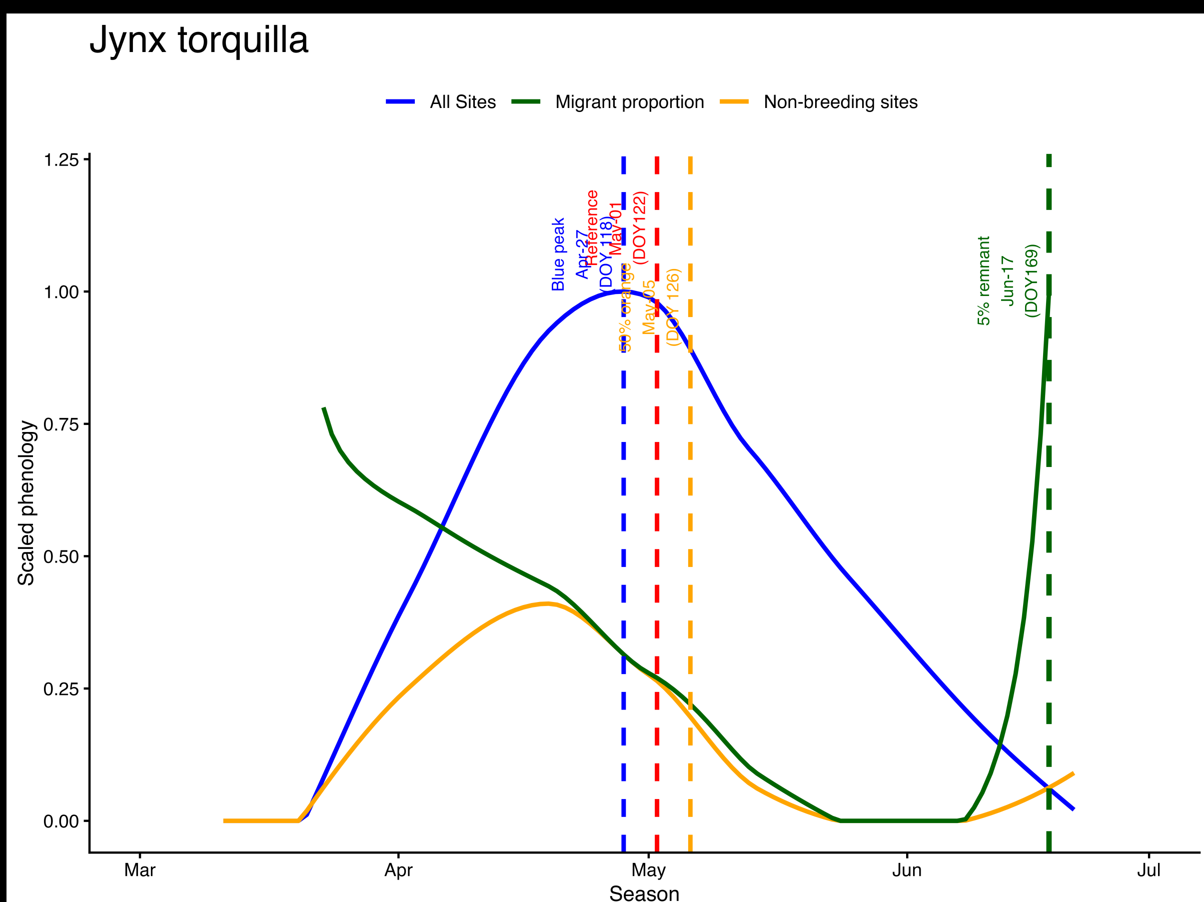


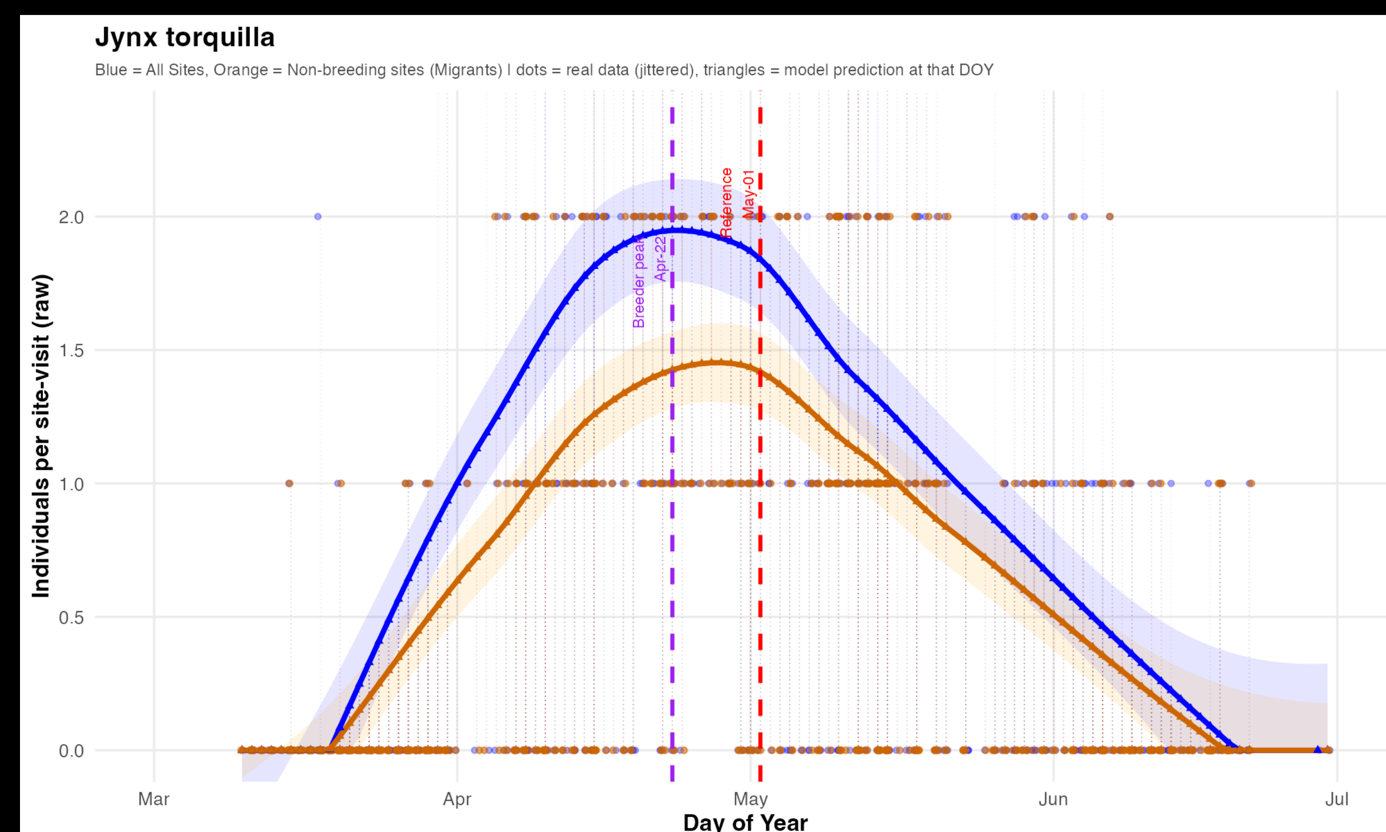


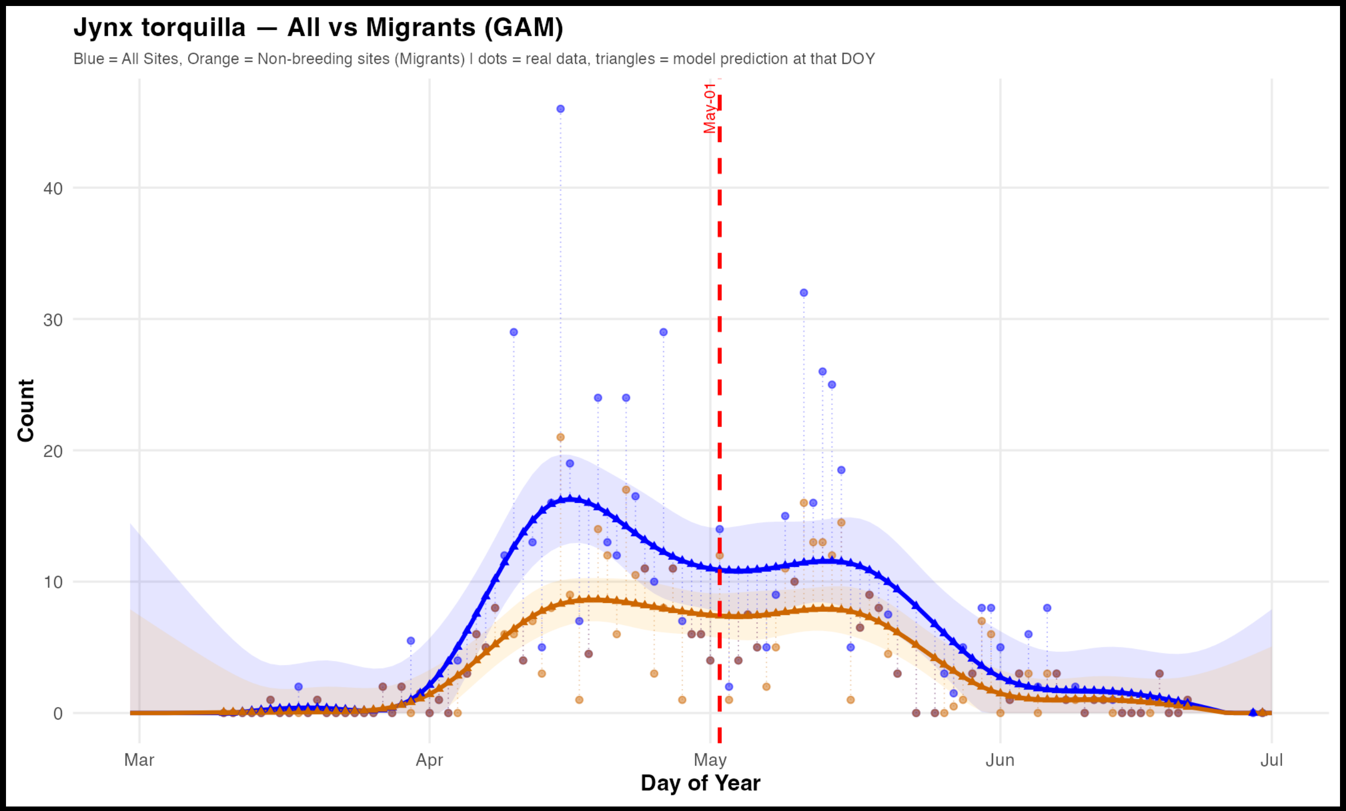


Bavaria,
Württemberg-I and II



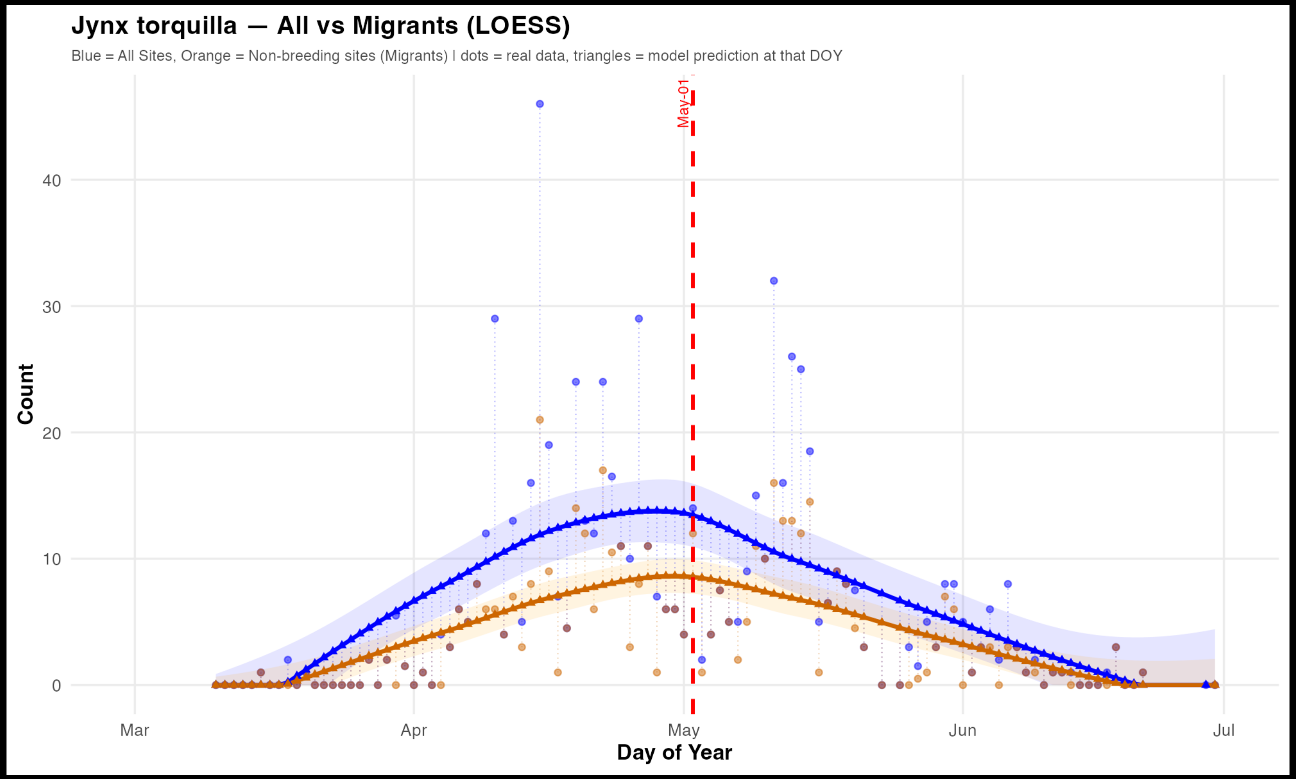




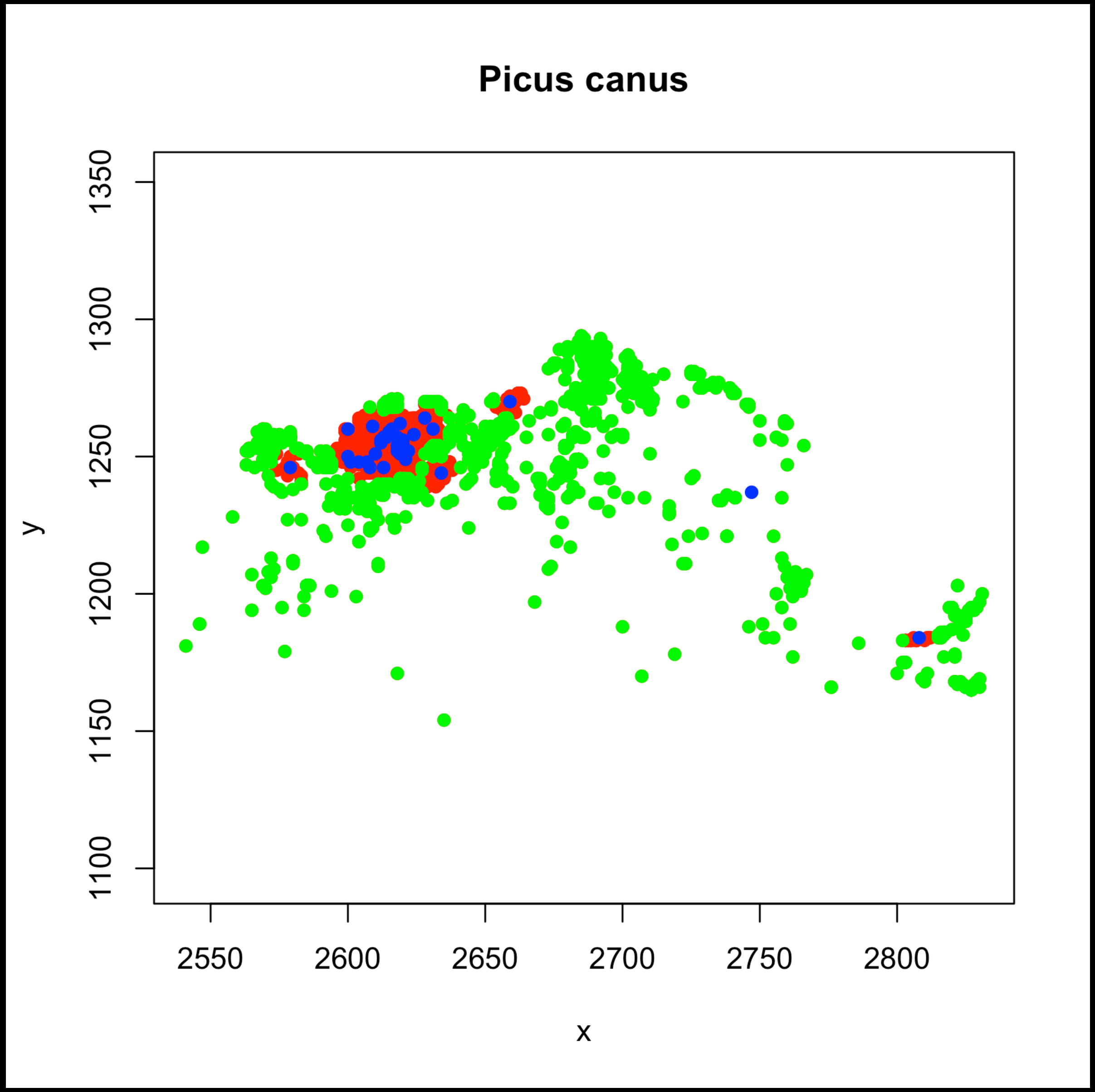


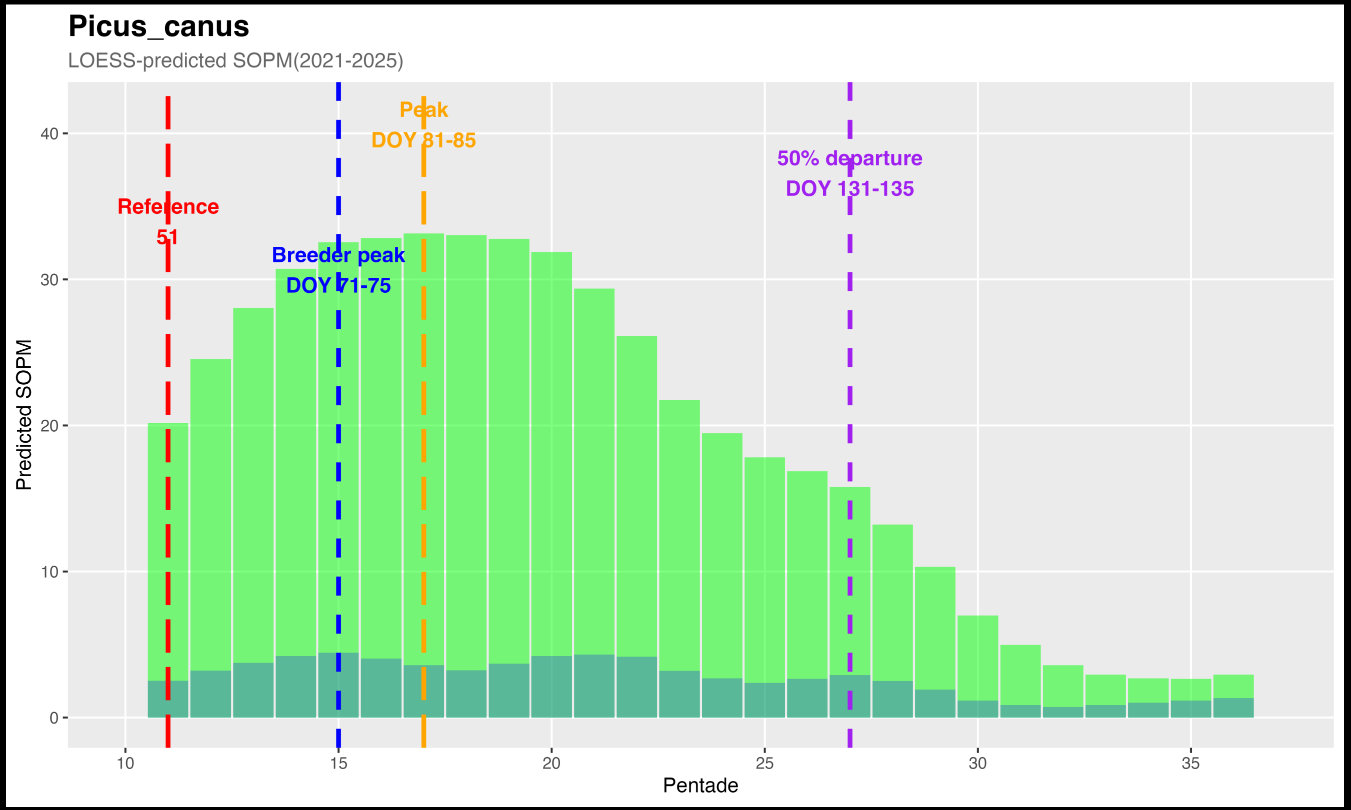
slight detour

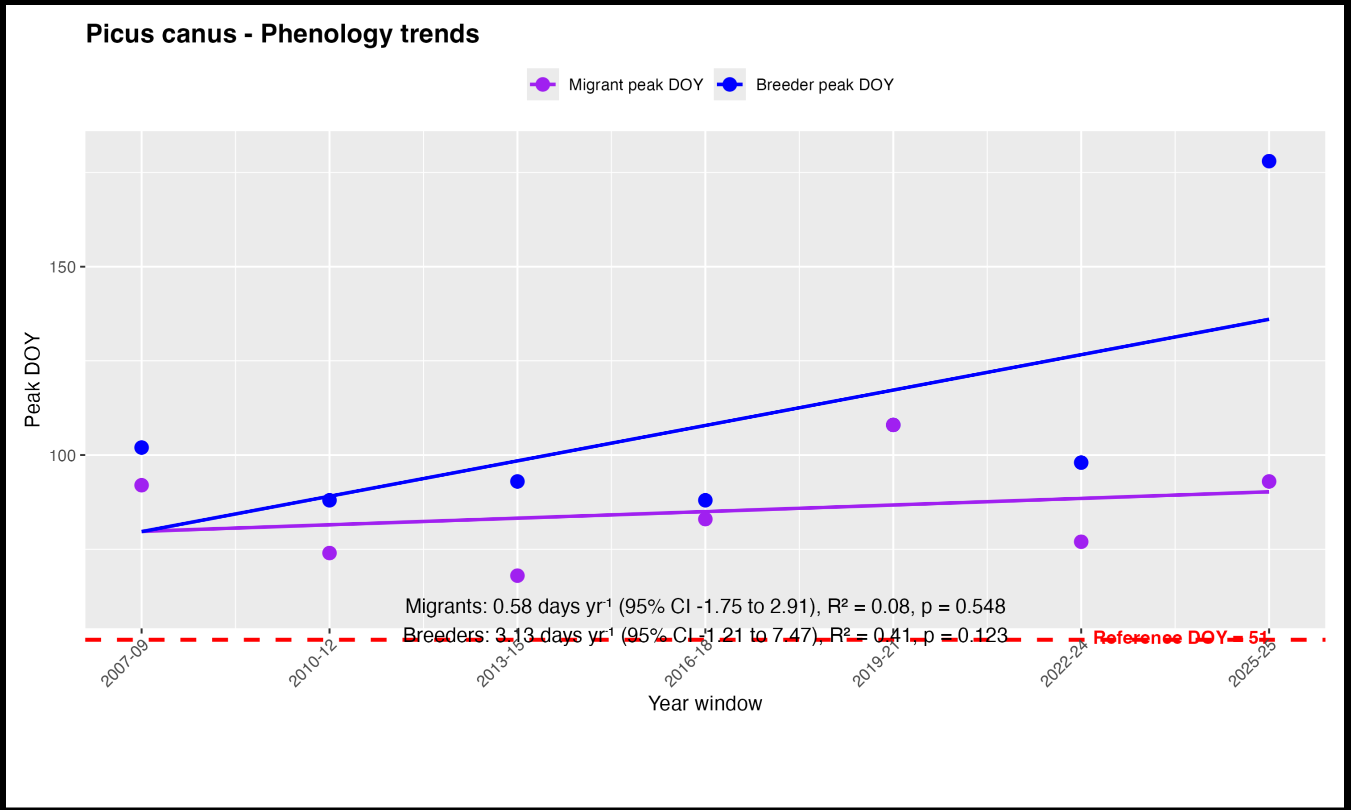
loess vs gam for
the same data



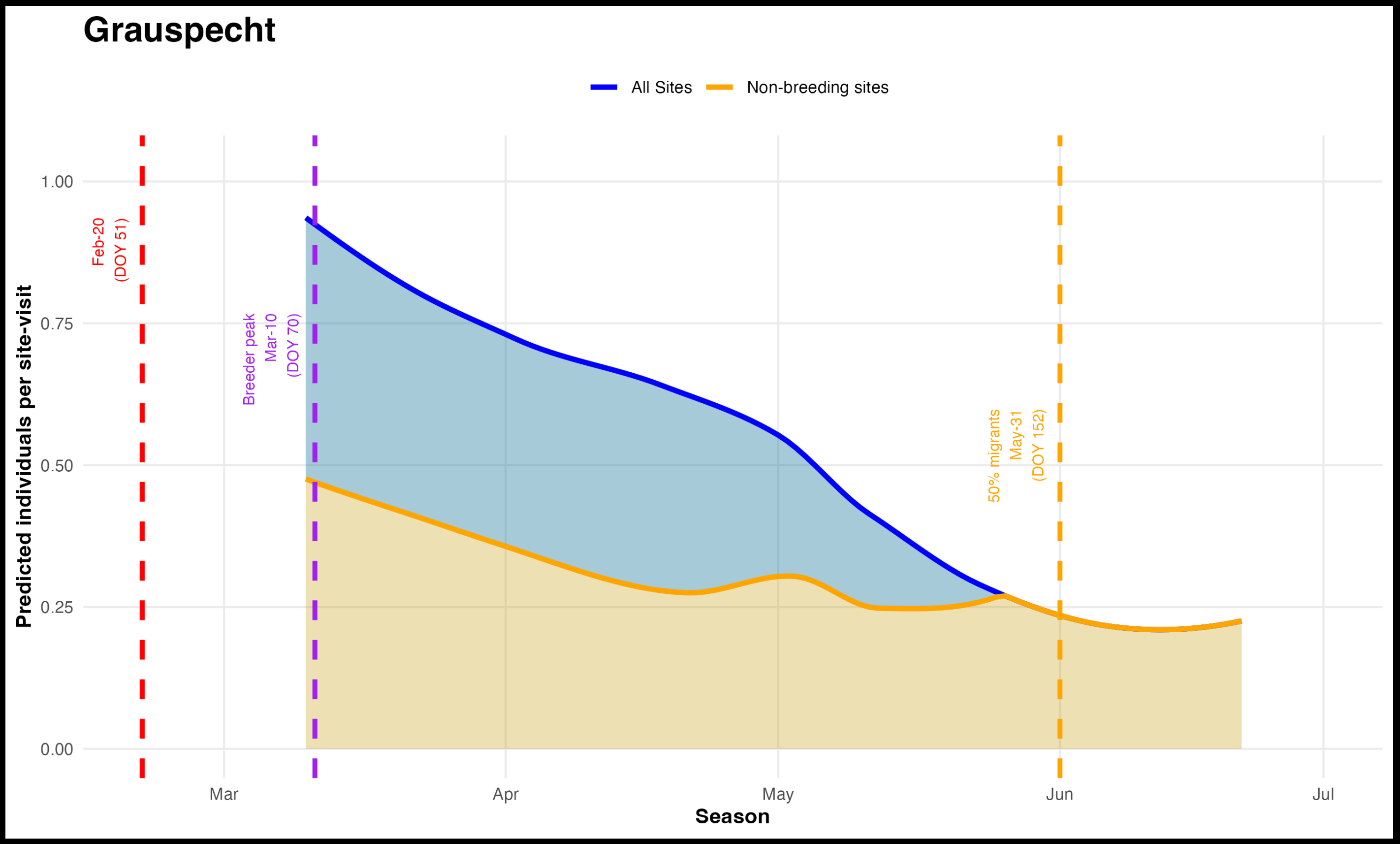
short distance migrants:
Picus canus

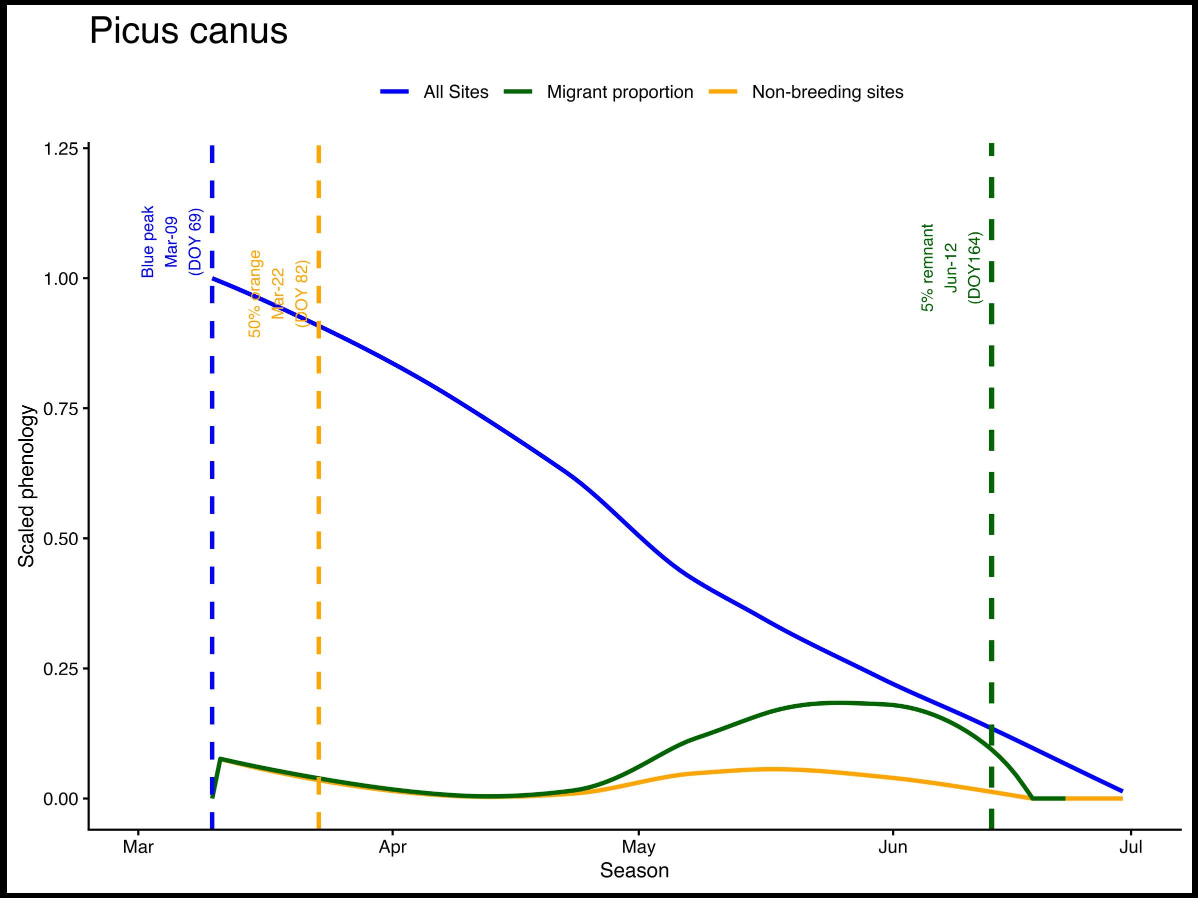






region-dependent abundance of the non-breeding counts





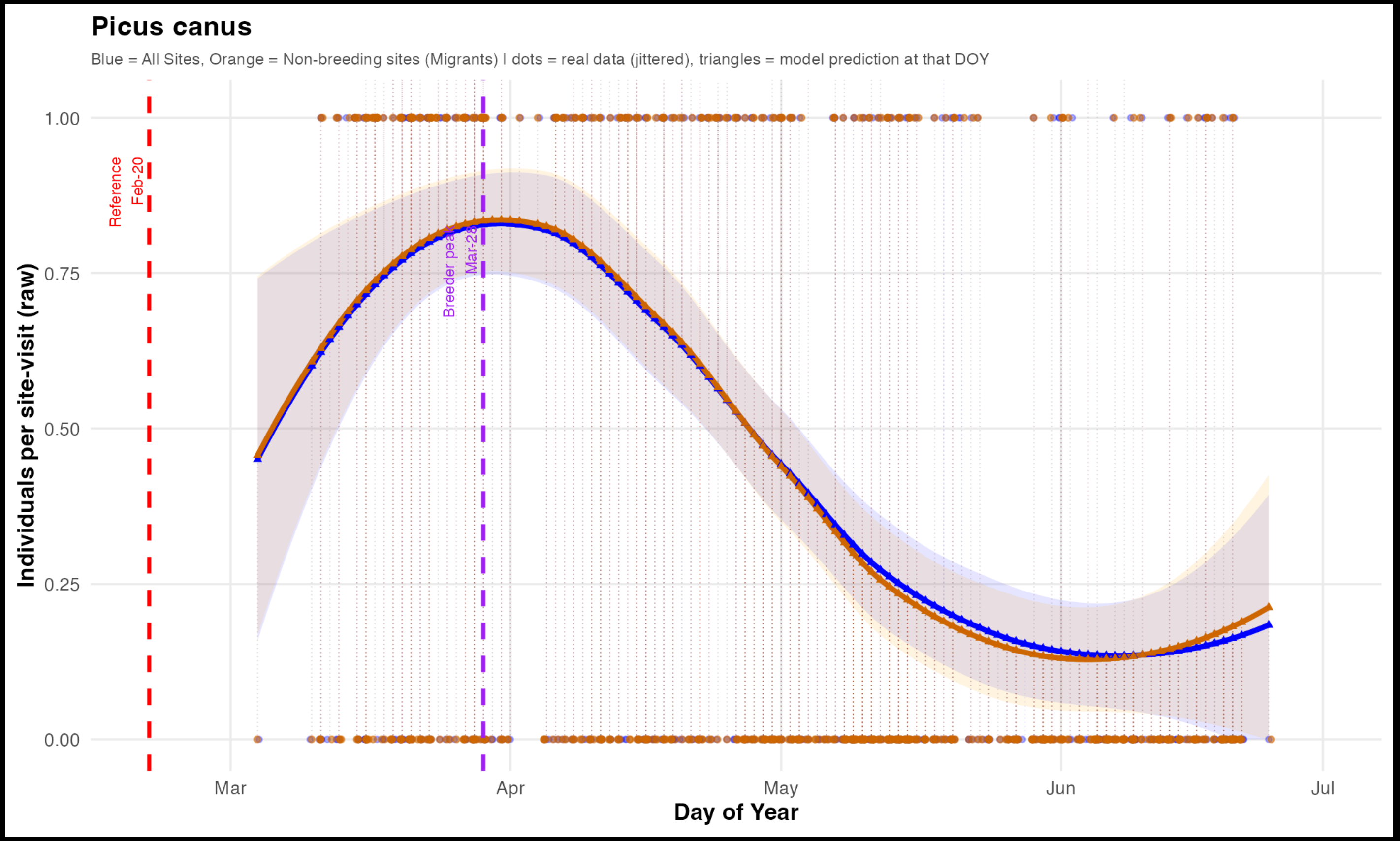
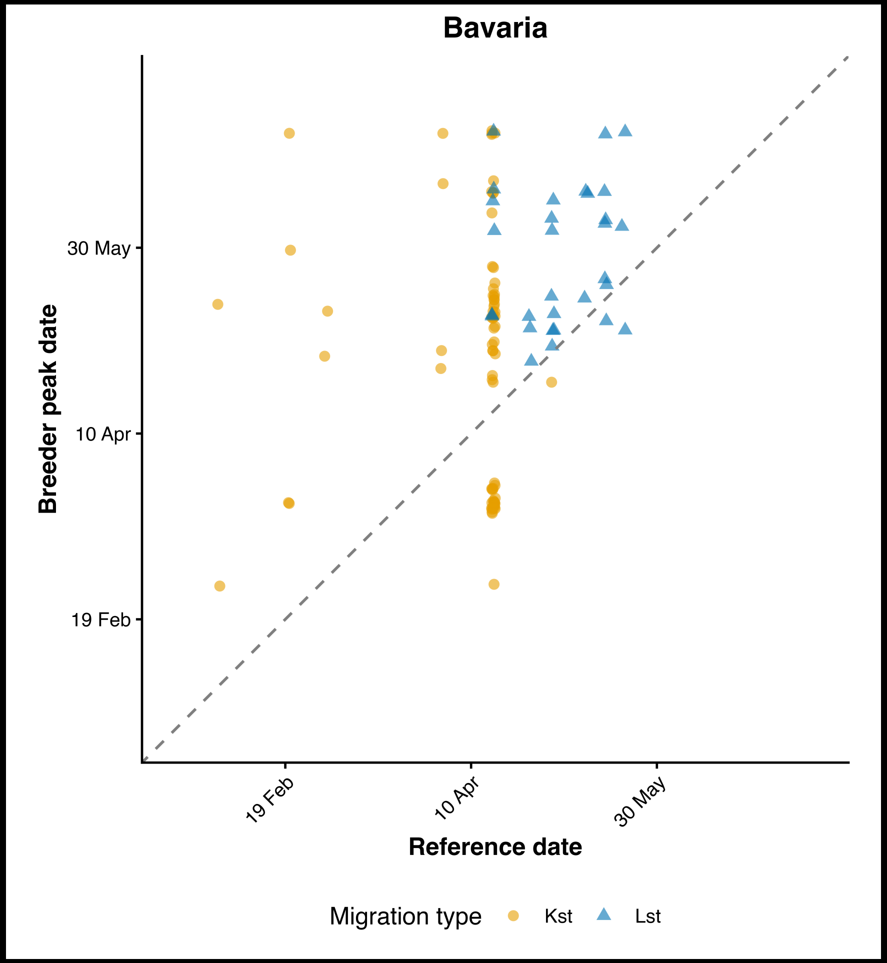
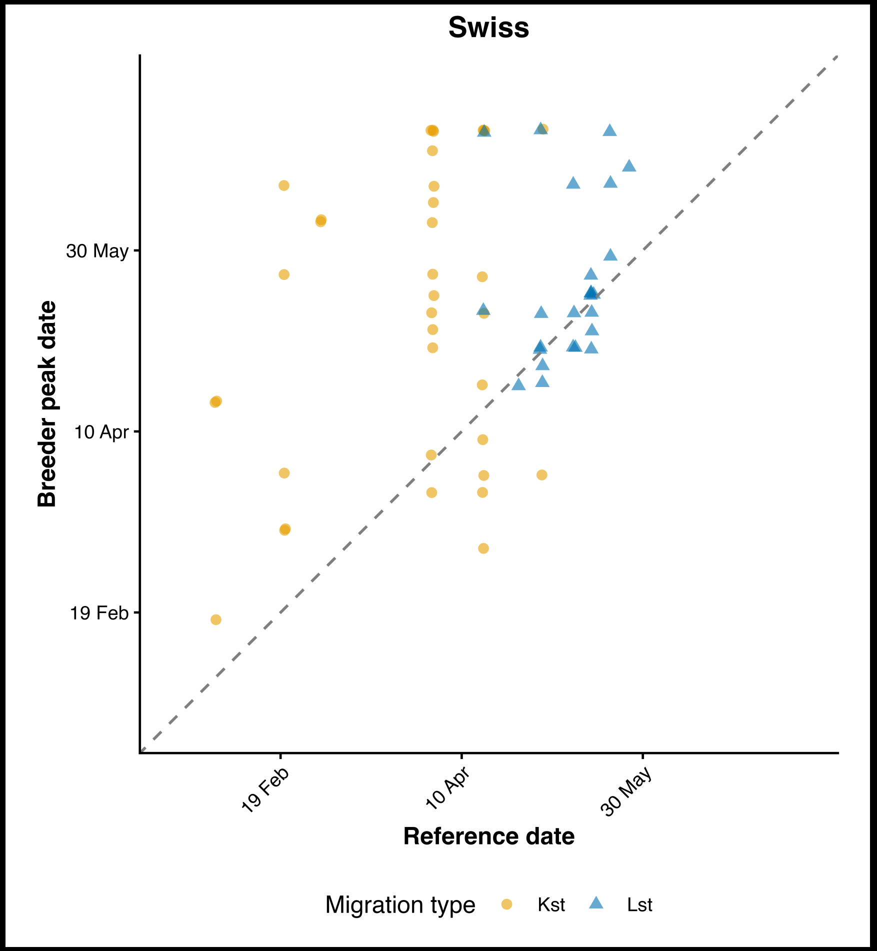


Table 5: Median (IQR) of the three metrics, in days, by dataset and migratory strategy. Negative = RD precedes the observed event and vice-versa. %neg = percentage of species in the distribution, yielding a negative response

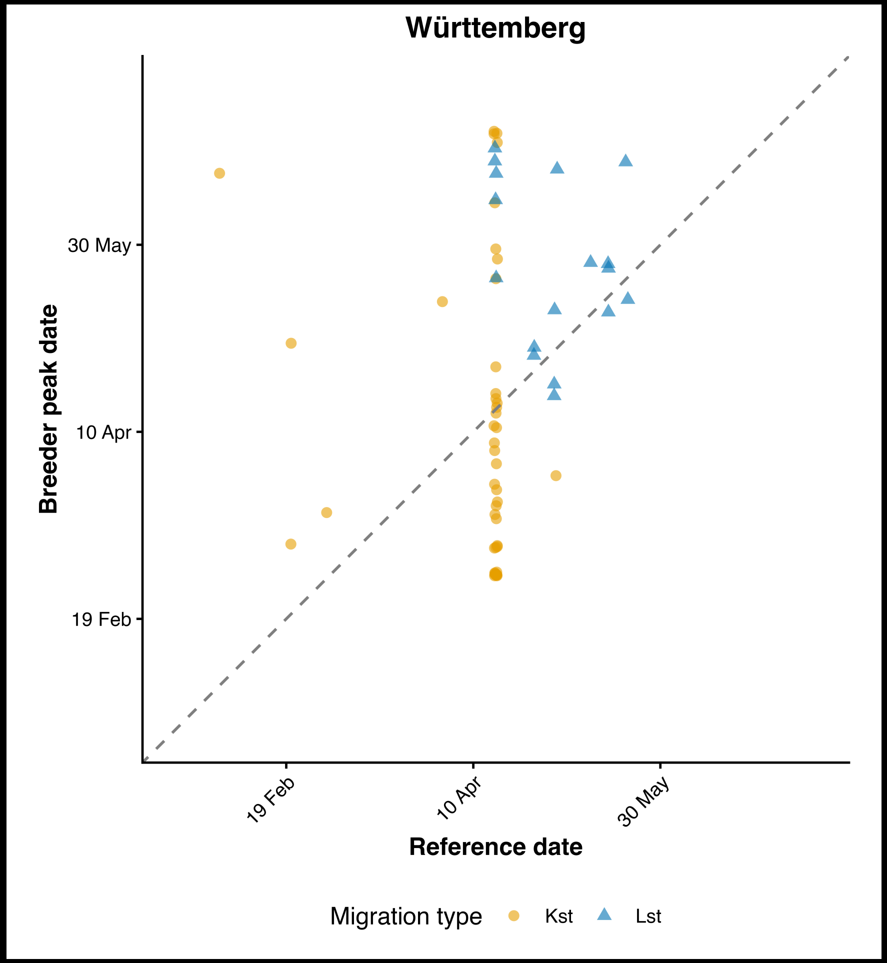
		Ref – breeder peak		Ref – migrant peak		Ref – migrant 50%	
Dataset	**Zug**	Med. [IQR]	%neg	Med. [IQR]	%neg	Med. [IQR]	%neg
Bavaria	Kst	−25 [−34, 23]	64.5	15 [−40, 27]	42.1	−12 [−55, 11]	62.5
Bavaria	Lst	−21 [−36, −6]	93.3	−5 [−33, 4]	51.6	−15 [−38, −7]	88.0
Swiss migration	Kst	−39 [−71, −6]	77.8	1 [−33, 28]	50.0	−51 [−71, −17]	89.7
Swiss migration	Lst	−2 [−25, 4]	61.5	3 [−11, 8]	46.2	−27 [−32, −11]	85.7
BW I	Kst	11 [−19, 37]	40.0	3 [−40, 37]	48.9	−13 [−61, 9]	54.8
BW I	Lst	−11 [−48, −4]	76.5	−20 [−40, 6]	58.8	−14 [−23, −10]	87.5
BW II	Kst	5 [−38, 37]	46.0	5 [−38, 37]	47.6	−28 [−68, −3]	84.4
BW II	Lst	−14 [−25, −2]	77.3	−15 [−31, −3]	81.8	−31 [−44, −29]	90.9

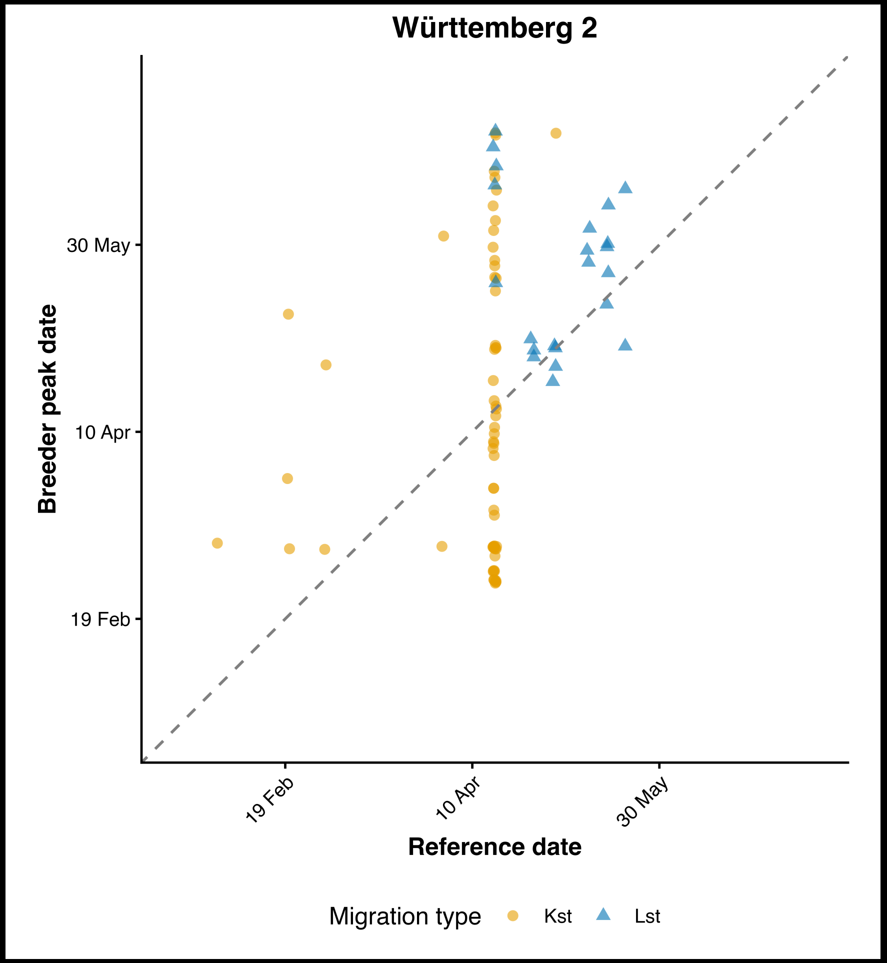
how do the breeder and migrant phenologies compare with the current reference dates?

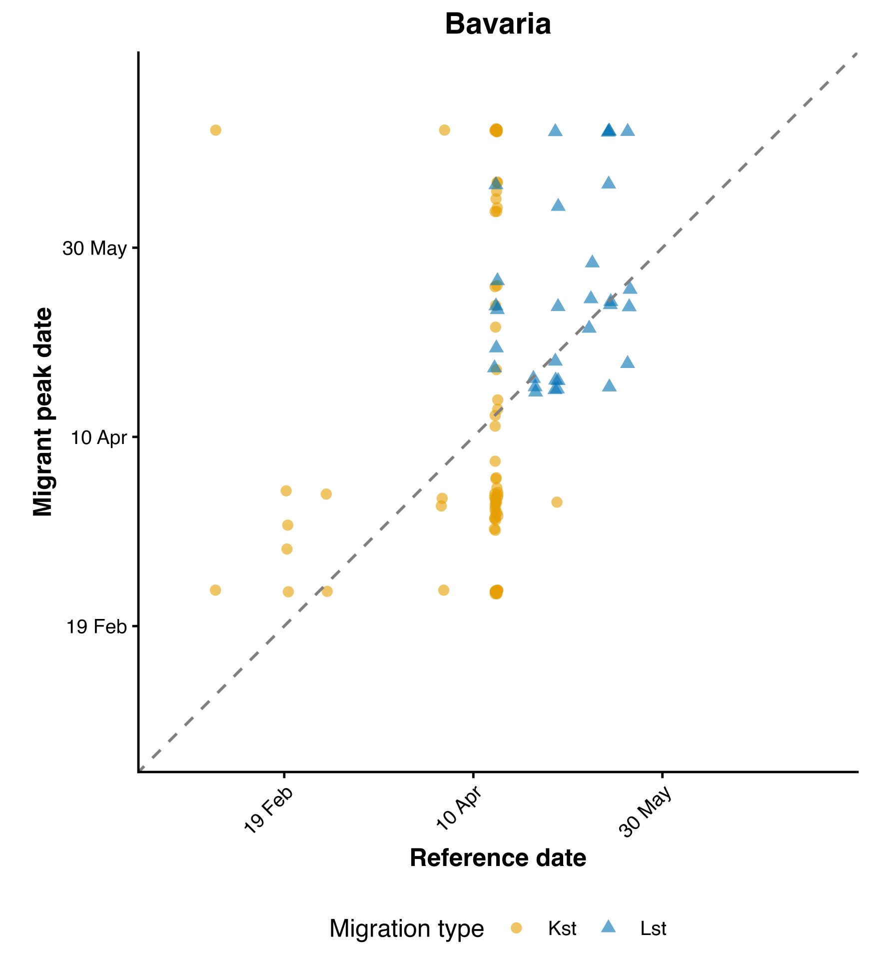

Bavaria
Breeder peak date
30 May
10 Apr
19 Feb
19 Feb
10 Apr
30 May
Reference date
Migration type
Kst
Lst

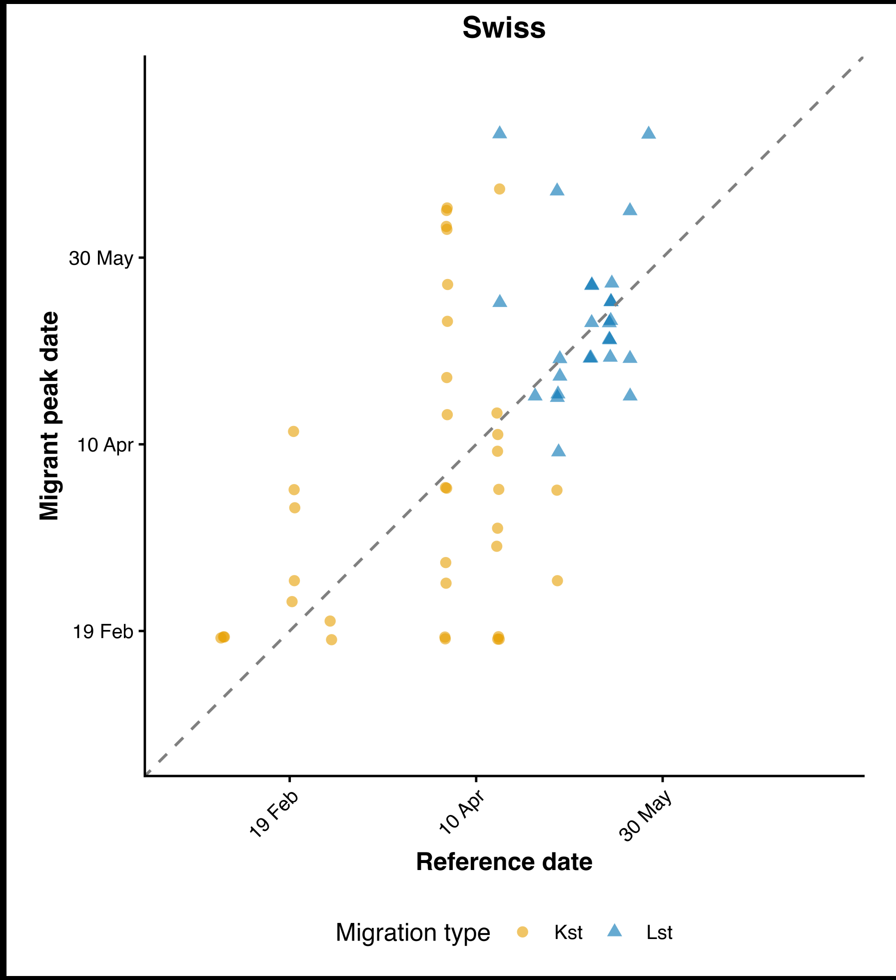

Swiss
Breeder peak date
30 May
10 Apr
19 Feb
19 Feb
10 Apr
30 May
Reference date
Migration type
Kst
Lst

q-i

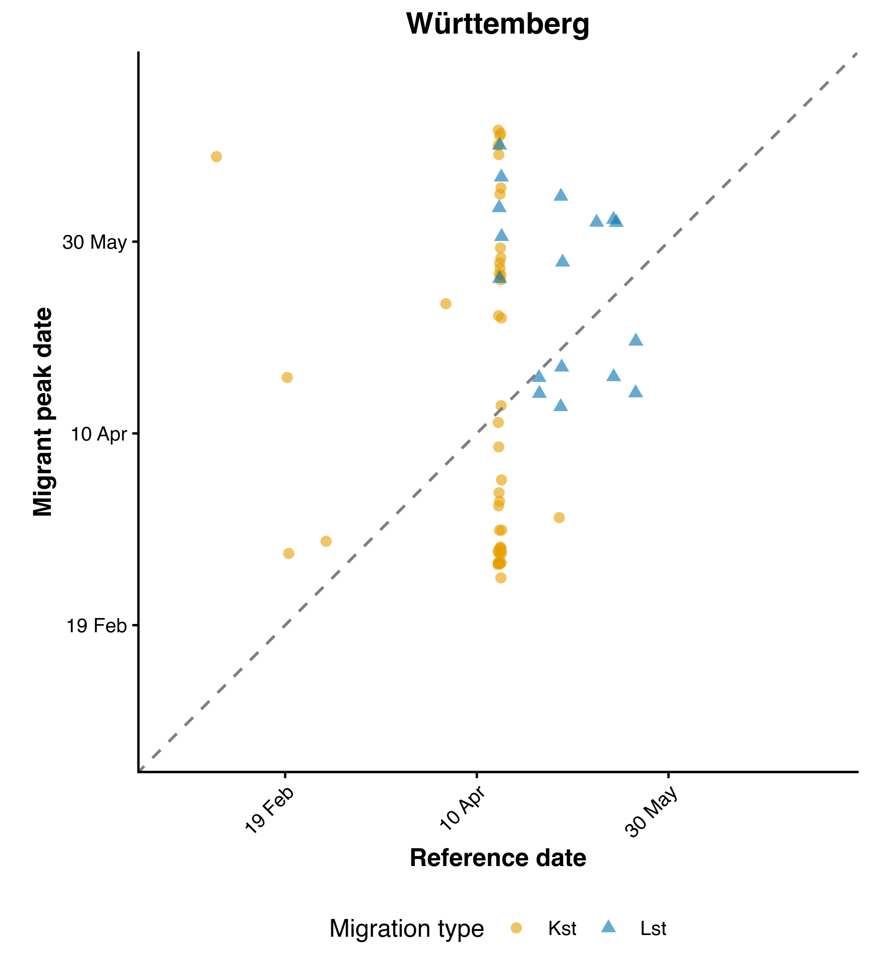

Württemberg
Breeder peak date
30 May
10 Apr
19 Feb
19 Feb
10 Apr
30 May
Reference date
Migration type
Kst
Lst

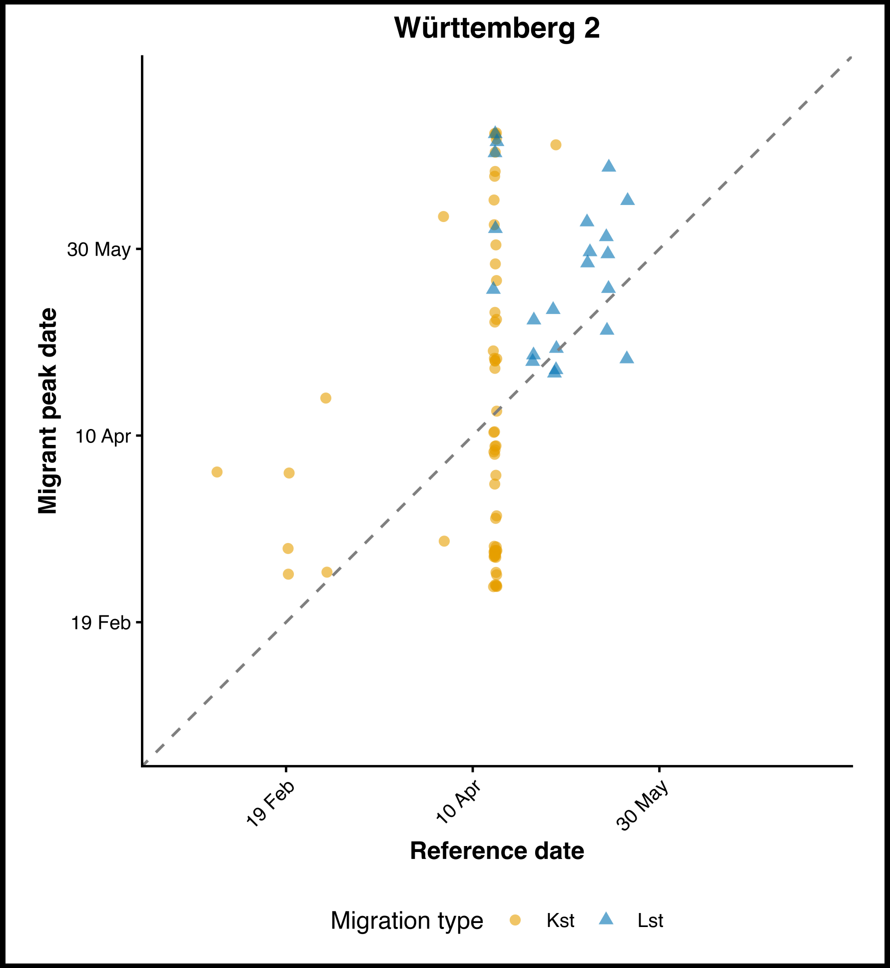

Württemberg 2
Breeder peak date
30 May
10 Apr
19 Feb
19 Feb
10 Apr
30 May
Reference date
Migration type
Kst
Lst


Bavaria
Migrant peak date
30 May
10 Apr
19 Feb
Reference date
Migration type
Kst
Lst


Swiss
Migrant peak date
30 May
10 Apr
19 Feb
Reference date
Migration type
Kst
Lst

q-ii


Württemberg
Migrant peak date
30 May
10 Apr
19 Feb
Reference date
Migration type
Kst
Lst

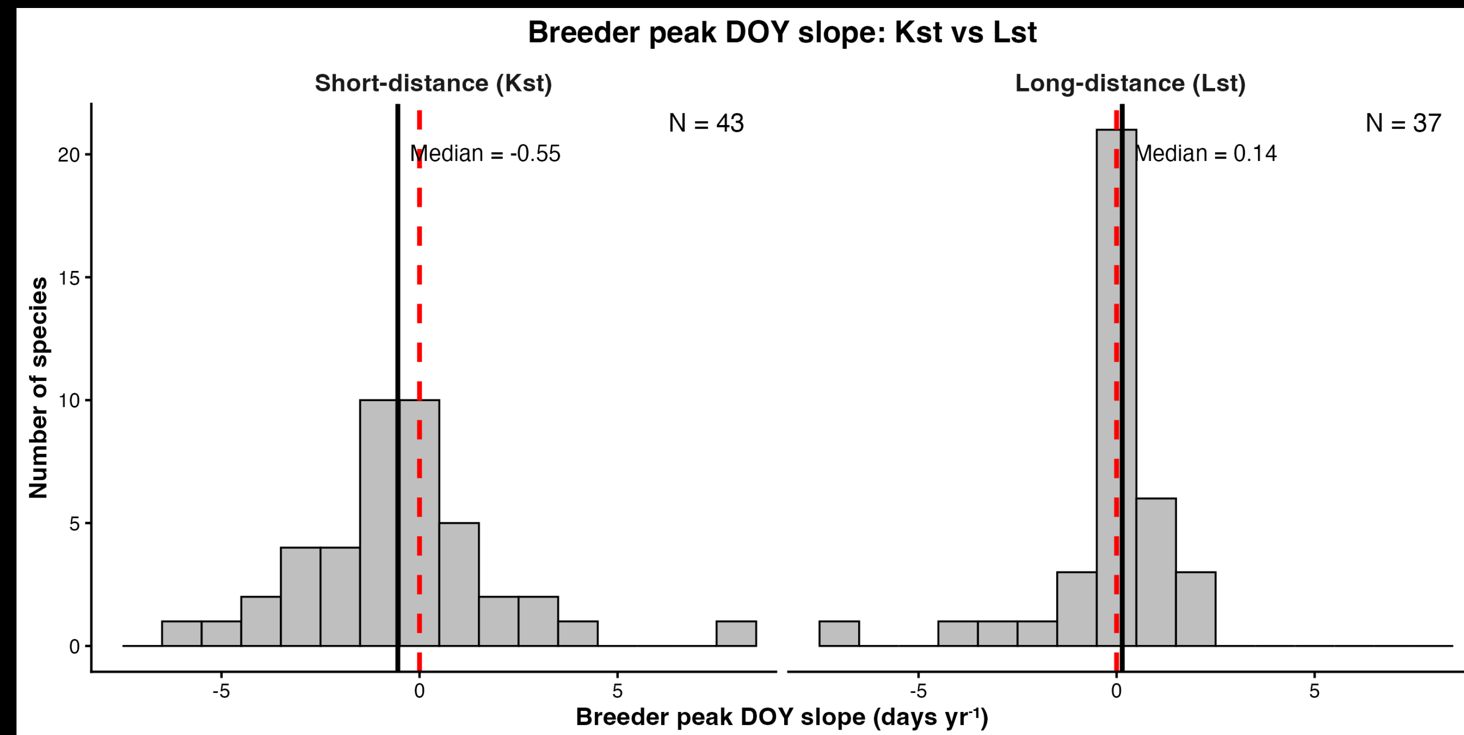

Württemberg 2
Migrant peak date
30 May
10 Apr
19 Feb
Reference date
Migration type
Kst
Lst

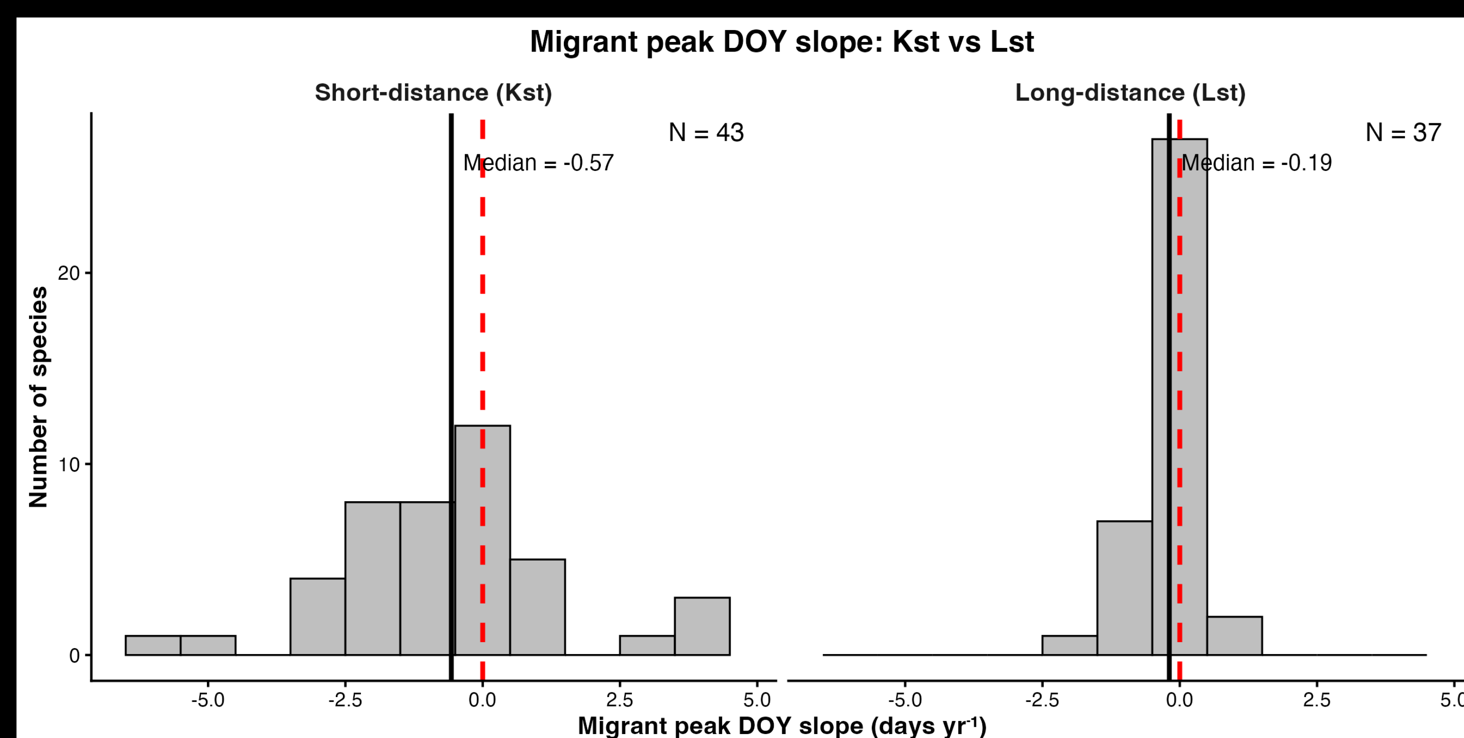
q-iii:

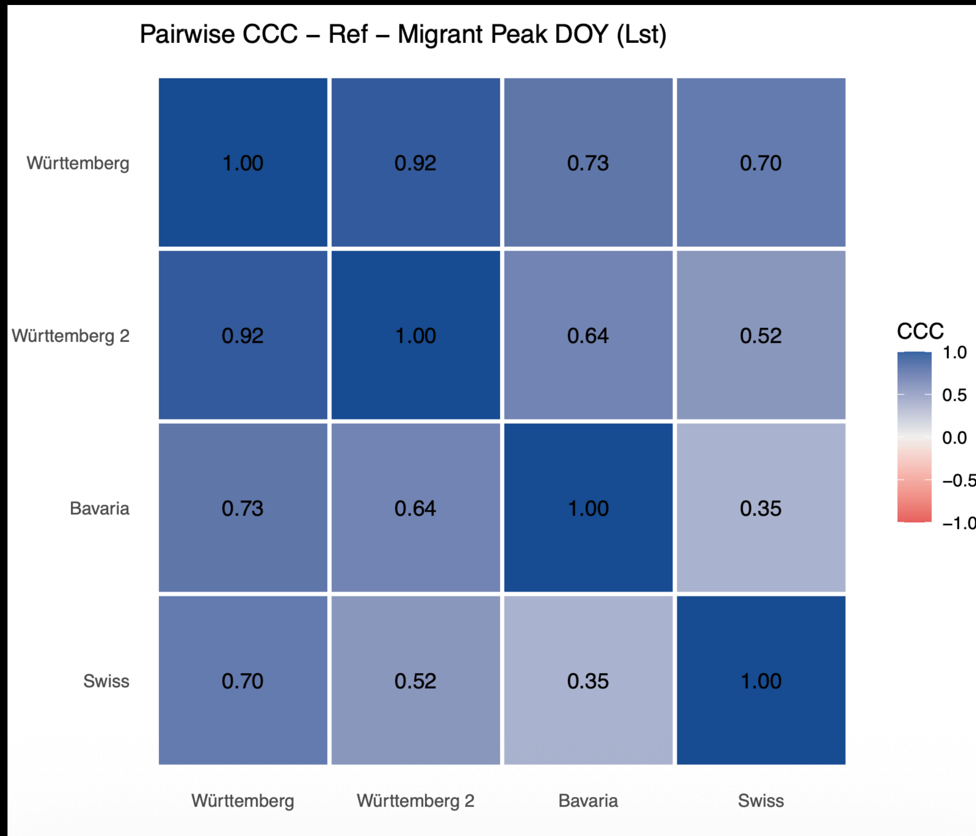
changes in migrants and breeders' peak arrivals over the years

Table 6: Swiss 2007-2025 trend slopes of metrics (days yr^{-1}), median [IQR], by migratory strategy; %neg = advancing; %sig = statistical significance at $p < 0.05$.

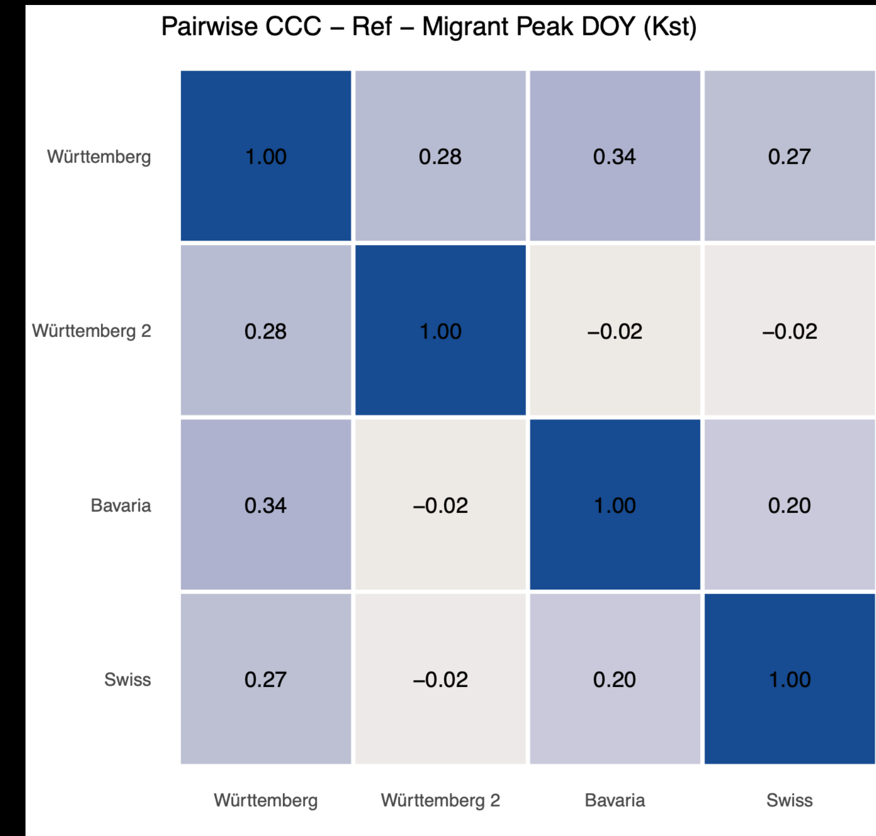
Metric / Zug	N	Median [IQR]	%neg (advancing)	%sig
Breeder peak – Kst	43	−0.55 [−1.69, 0.55]	60.5	11.6
Breeder peak – Lst	36	0.15 [−0.13, 0.47]	38.9	5.6
Migrant peak – Kst	43	−0.57 [−1.77, 0.28]	67.4	23.3
Migrant peak – Lst	37	−0.19 [−0.44, 0.01]	70.3	5.4

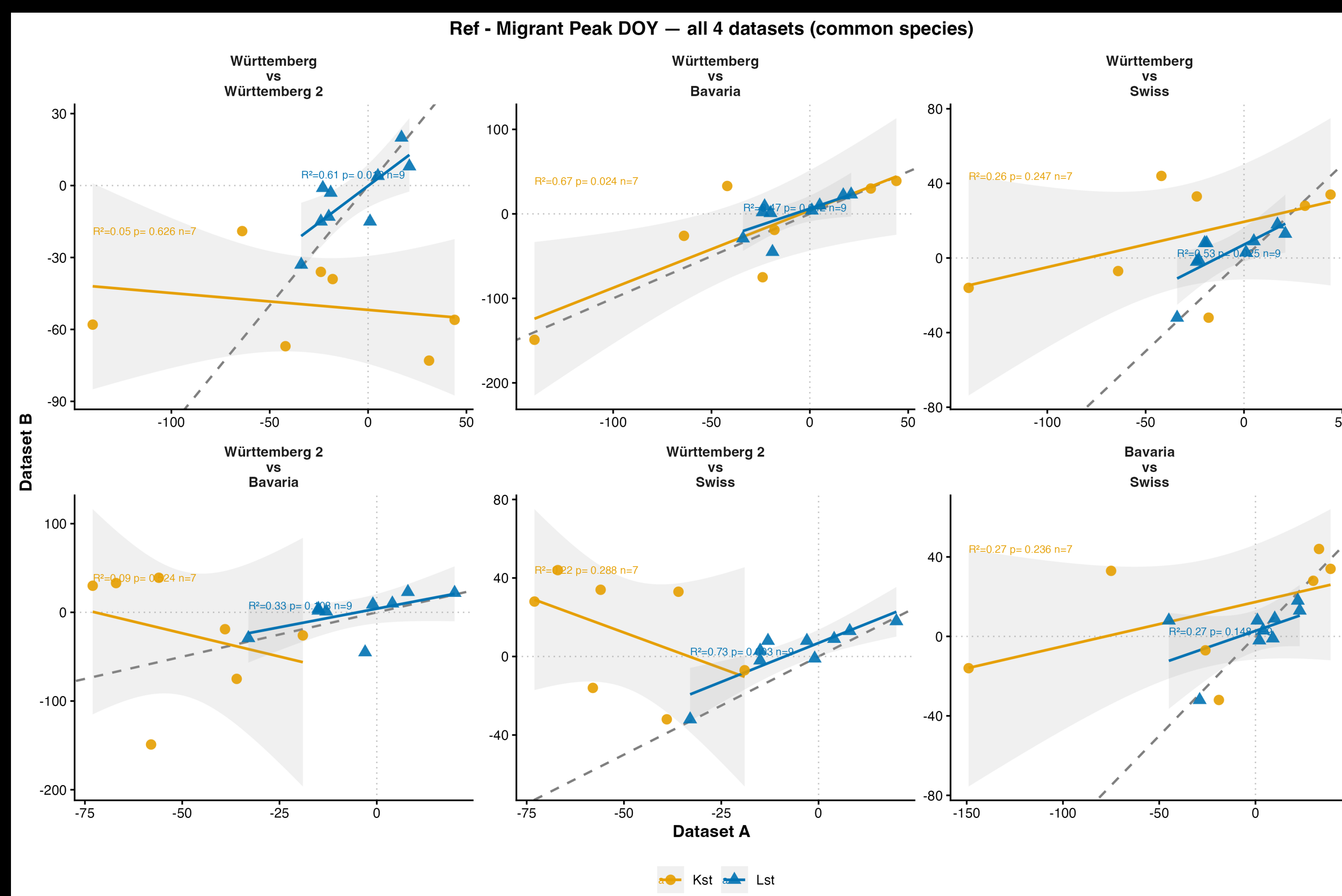






what does cross dataset agreement look like?





(disagreement, as well)

Table 7: Median pairwise Pearson correlation (r) across the six pairwise dataset comparisons, by metric and migratory strategy

Metric	Kst	Lst
Migrant peak DOY	0.07	0.65
Breeder peak DOY	0.46	0.51
Ref − migrant peak	0.28	0.74
Ref − breeder peak	0.59	0.66